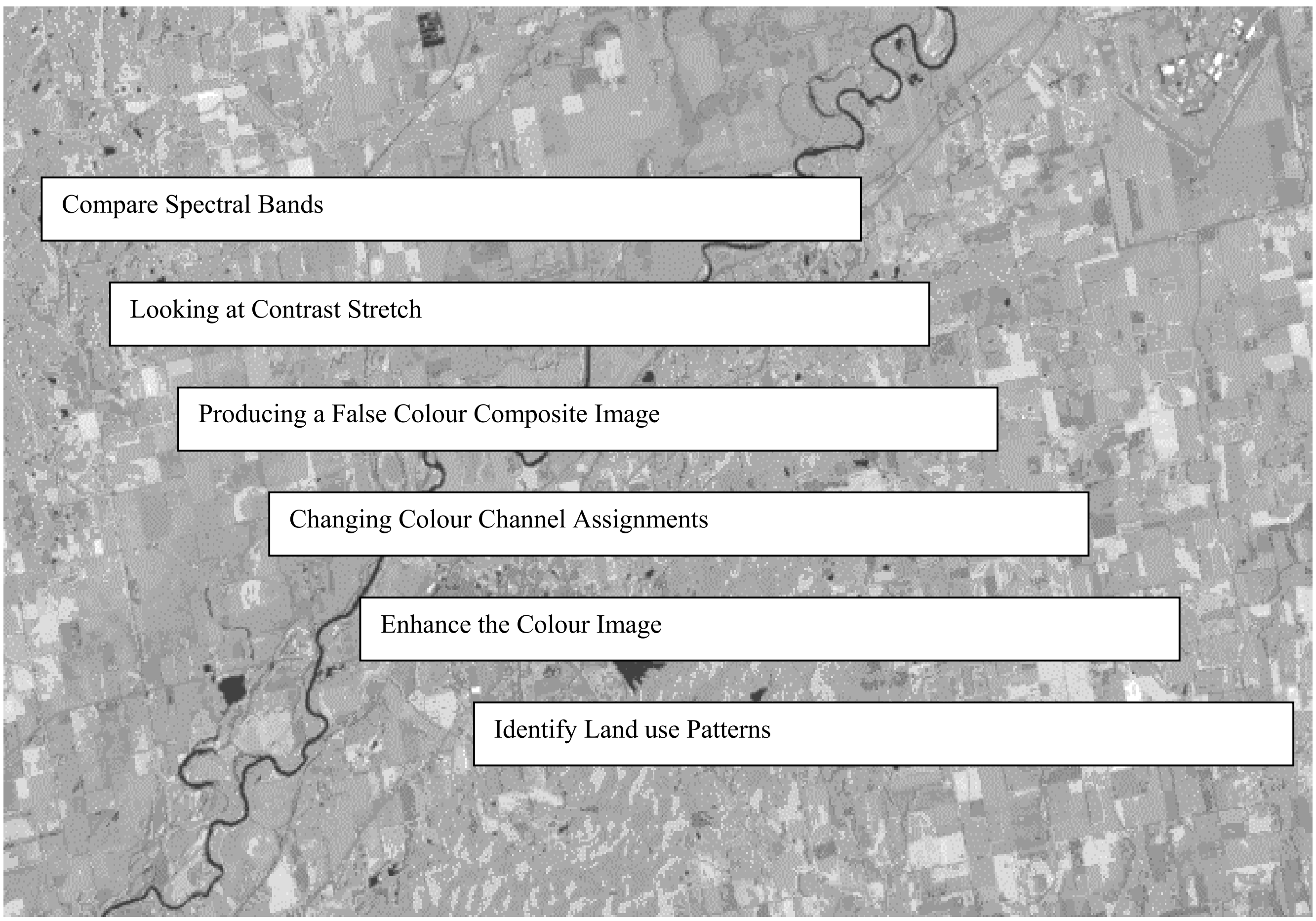
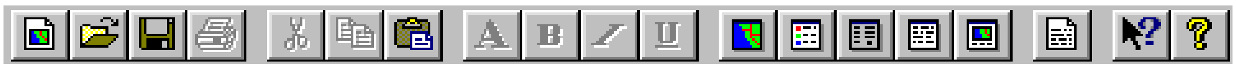
Land Use and Land Cover of Canada



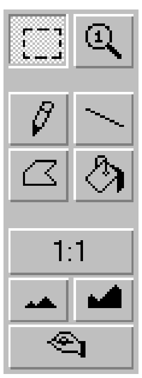
MFworks Lesson

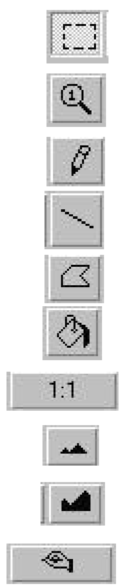
MFWorks Icons and Tool Bars

Create new map	Open map	Save Map	Print Map		Cut	Copy	Paste		Font	Bold	Italic	Underline		View Legend	Select Legend	Display Information	Display Comment	Display Layout		New Script		Help	About



Tool Bar





Select Tools
Numeric Magnifying Glass
Freehand Drawing Tool
Draw Line Tool
Draw Polygon Tool
Fill Tool
Zoom To
Zoom Out
Zoom In
Redraw Map

For more information check: http://www.thinkspace.com

STARTING MFworks

Before you begin, the lesson data folder must be ***copied*** into a drive in which the student has writing and editing capabilities. This will probably be the drive containing your own personal account on the school's system or it may be the **C: drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete. Each school set-up is different so work with your teacher to prepare your data folder.

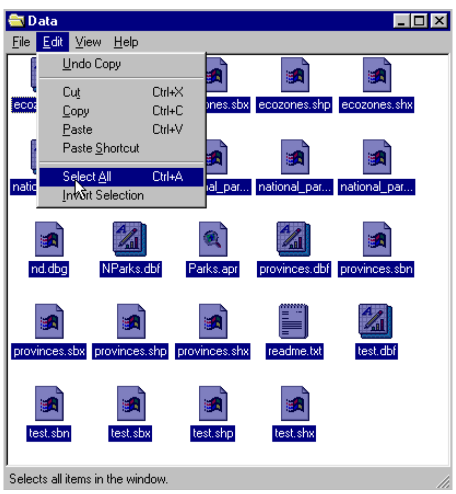
! *(The data for the lesson must first be downloaded from this Web site and placed on the school computer or network. Ask your teacher where to find the data folder that has been downloaded).*

To do this, use the file manager program on your computer. For windows based systems this will be **My Computer / Windows Explorer /.**

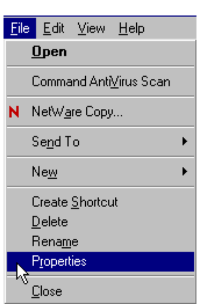
Once the data folder has been ***copied*** to the required destination the files must be made usable by removing the ***read only*** attribute. For windows based systems follow these instructions:

! *(The data downloaded from the Web Site is set as **read only** so that it cannot be changed or altered by accident.)*

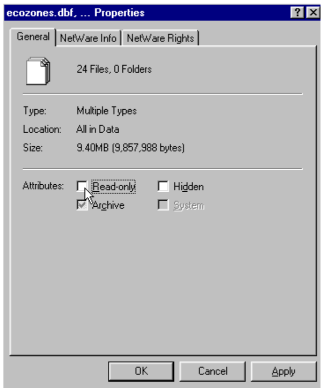
1. Open your new data folder
2. Choose **Edit** > ***Select All*** (they will all appear Blue)



3. From the menu choose:
   File > Properties



4. Remove the check mark in the box beside the label ***read only***.



5. Click **Apply**, then **OK**

The data in your new data folder can now be opened, edited and saved using MF Works to carry out the following project.

! Denotes an observation.

? Denotes a question that may be answered on the Student Activity Sheet.

Introduction to Remote Sensing

This lesson will deal with raster images. It will investigate how a raster image is produced, the various factors that affect the image and how we can work with it to define land use on the ground.

First, here are some facts about where the image comes from and what produces the image we get.

THE SATELLITE and SENSOR

Landsat 5 satellite
Thematic Mapper Sensor
More information on this satellite and how it works can be found at http://www.ccrs.nrcan.gc.ca/
Or more precisely to their Education section at:
http://www.ccrs.nrcan.gc.ca/ccrs/eduref/edurefe.html

Also, free Landsat 5 TM Data can be found at www.geogratis.gc.ca

The Thematic Mapper sensor records images of the earth's surface in seven different spectral bands at the same time thus the term multispectral. A spectral band is selected part of the electromagnetic spectrum. The visible light we see with our eyes is only part of the electromagnetic spectrum. By recording images in different spectral bands the Thematic Mapper can show us many different aspects of the earth's surface that we may not see with in a normal single band image. The following list presents the colour and wavelength of the seven spectral bands recorded by the Thematic Mapper.

Multispectral bands	Wavelength (nanometers)
TM1 Blue	0.45-0.52 μm
TM2 Green	0.52-0.60 μm
TM3 Red	0.63-0.69 μm
TM4 Near Infrared	0.76-0.90 μm
TM5 Mid-infrared	1.55-1.75 μm
TM7 Mid-infrared	2.08-2.35 μm
TM6 Thermal	10.4-12.4 μm (not used in this study)

30 m spatial resolution (cell size is 30 m X 30 m)
Repeat coverage: 16 days at equator
185 km swath (sub-scene 446 x 726 cells)
Number of radiometric levels: 256 (8-bit)
Date our image was acquired: May 29, 1994

CHARACTERISTICS of our study area.
These are the features that we can expect to see in the image we will use in this study.

Definitions:

Bedrock: shales in the east, sandstone hills to the south and limestone hills to the west with sinkhole present.
Soils: parent materials are shale, limestone and sandstone. In general the soils are fine loam or loamy clay with a reddish colour. Along the major drainage channels the soils are more clay and have a chocolate-colored topsoil.

Land cover; crops, pasture, hardwood forests (Oak-Hickory association), wood lots, water, towns, roads and airport.
Land use: numerous small family farms; grazing cattle and raising horses; built-up areas; transportation; forestry, and water control.
Hydrology; meandering river, oxbow lakes, sink holes, manmade reservoirs and ponds, abandoned river channels.

<u>Begin the lesson by starting up the MFworks program.</u>

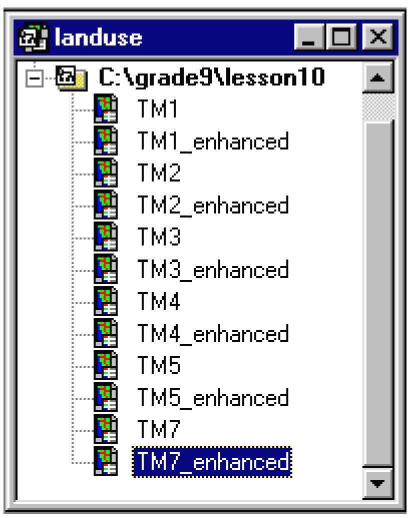
1. From your system Start menu, choose **Programs > MFworks > MFworks.**

The MFworks program will launch. If necessary, click the window maximize button so that the MFworks window fills the screen.

! *When the program starts, it automatically opens a window titled "**Untitled Project**." MFworks uses **projects** to group together and organize maps that are used together. A project has been created for you to use with this activity, so you do not need the default project.*

2. Choose **File > Close** to close the Untitled Project window.
3. Choose **File > Open Project** to bring up the Open dialog box.
4. Make sure that the folder named in the "**Look in**" field is the data folder just created.
5. Select the file name **Landuse.mfp** and click the Open button to open that project.

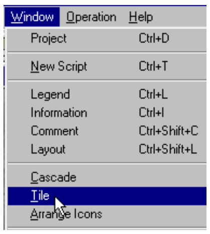
! *The* **Landuse.mfp** *project lists twelve maps that you will use in this activity. You will also create several others using MFworks operations. As they are created, these new maps will automatically be listed in the Project window. In order to keep the workspace usable it is important to close or minimize any maps not in use leaving only one map open on the screen to work on. If you close a map ignore the "Save Changes" message where appropriate.*



This lesson looks at we can use multispectral remote sensing data in MFWorks. We will compare the thematic content of different spectral bands and prepare enhancements to improve interpretation. We will also combine several spectral bands to produce colour composites for our study area, which can improve interpretation of land use and land cover.

COMPARE SPECTRAL BANDS

1. Double click **TM1** to open the blue spectral band.



2. Click and Drag the edge or corner of the "**TM1**" map window to enlarge the Map window to fill your screen or maximize the image window.
3. Click MENU item **Window > Landuse.**
4. Double click **TM4** to open the near-infrared spectral band.
5. Click MENU item **Window > Tile**. This will show all the open windows on the screen at one time.
6. Close the two **Legend** windows and minimize the project window.
7. Click **Window** > **Tile** again.
8. The images for **TM1** and **TM4** should both be visible on the computer screen.
9. Scroll through these two images and compare.

Things to remember:

1. each spectral band measures reflected light for specific parts of the colour spectrum, e.g. band 1 is blue light; band 2 is green; band 3 is red; band 4 is near-infrared; band 5 and 7 are mid-infrared; and band 6 is thermal infrared.
2. dark shades represent low reflectance of light
3. light shades represent high reflectance of light
4. green plants with chlorophyll have high reflectance in the near-infrared (seen as bright areas)
5. water has very low reflectance in the near-infrared (seen as dark areas)
6. bare soil and concrete have relatively high reflectance of blue light and other visible light (seen as bright areas)
7. A cell is 30 meters X 30 meters of the image and represents a path that is 21.78 kilometres in width.

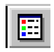
? **What features do you see?**

? **What are some differences between the two spectral bands?**

! **Open and look at some of the other TM bands. Compare them to TM1 and TM4. Close all but TM1 and TM4 when you are done.**

LOOKING AT IMAGE ENHANCEMENT

We will now look at ways that the data in the spectral bands can be enhanced to increase the number of features that can be detected and improve interpretability.

1. Close the **TM4** image window.
2. With only the **TM1** image open, maximize the window.
3. Click  **legend** button.

The first field of numbers represents pixel values ranging from 0 (black) to 255 (white). Note that the lowest pixel value found in the **TM1** image is 54 the highest is 255. The second field of numbers represents the number of cells with that value.

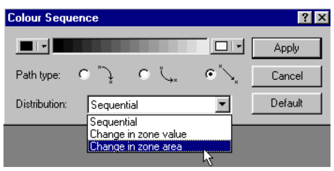
4. Scroll down the legend.

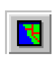
? Which cell value has the greatest number of cells?

? What can be said about the distribution of cell-by-cell values?

Enhancements of image data is done by distributing the cell values more evenly over the full range from 0 to 255.

5. With the legend **TM1** window active.
6. Click MENU item **Edit > Select All**.
7. Click MENU item **Legend > Colour Sequence**.



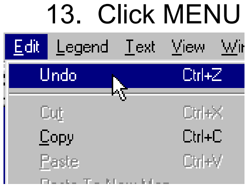
8. Open the **Distribution** list.
9. Click **Change In Zone Area**.
10. Click **Apply**.
11. Click the  **Map** button.

! Note the changes in the TM1 image.

? Is the enhanced image easier to interpret?

? Why?

12. Click the **legend** button.
13. Click MENU item **Edit > Undo**.



14. **TM1** should be back to the unenhanced state.

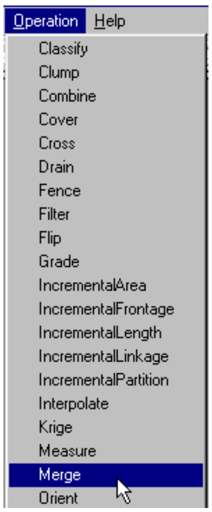
! Try a contrast stretch enhancement on a different spectral band by following the same procedure as above, replacing *TM1* with another *TM* band.

? **What effect does the enhancement have on this band?**

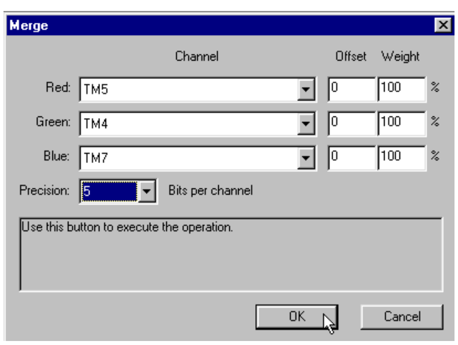
PRODUCING A FALSE COLOUR COMPOSITE IMAGE

We will now see how by putting several spectral bands together using different colours we can produce a colour composite that will enhance land use and land cover.

1. Close all the open image map layers.
2. Click MENU item **Operation > Merge**.



3. Open **Red Channel List > Select TM5**.
4. Select **TM4** for the ***Green*** **Channel**.
5. Select **TM7** for the ***Blue*** **Channel**.
6. Open **Precision List** > **Select 5**.
7. Click **O.K**.

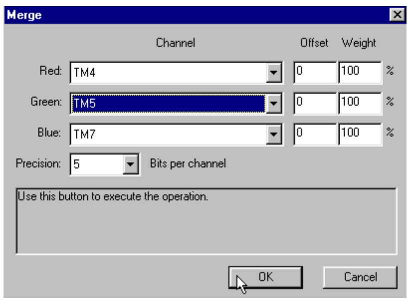


? **What type of image is produced?**
Describe some of the main image features.

CHANGING COLOUR CHANNEL ASSIGNMENTS

Now we will explore how using different spectral bands in a colour composite will change the look of the colour composite.

1. Click MENU item **Operation > Merge**.
2. Change **Red Channel** to **TM4**.
3. Change **Green Channel** to **TM5**.



4. Click **O.K.**
5. Click MENU item **Window > Tile**.

! Compare the two images.

? What differences do you see between the two images?

The second image produced simulates colour infrared photography in which plants with chlorophyll appear in shades of red (the more chlorophyll, the brighter the red), bare soils, rock and roads appear in shades of blue and green, and water appears black.

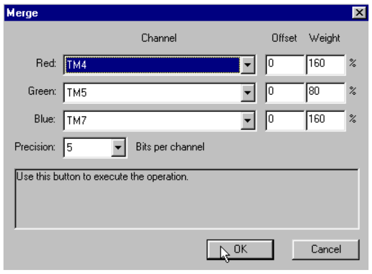
We will use this image (Map Layer 2) for our land use mapping. Close all other maps except this one.

ENHANCE THE COLOUR IMAGE

We will now look at enhancement of the colour composite that will improve the interpretability.

1. Click MENU item **operation > merge**.

2. In the **weight** field enter *160*
80
160



3. Click **O.K**.

? **How has the colour infrared image changed?**

IDENTIFY LAND USE PATTERNS

With the enhanced colour infrared image open, Click the **Layout**  button.

? **Find and label the following land use features.** Use the **Add Text**  tool to name these features.

1. **Airport**
2. **Town**
3. **River**
4. **Reservoir**
5. **Road**
6. **Forest**
7. **Woodlot**
8. **Cropland**
9. **Clay soils (appear blue)**
10. **Sandy soils (appear greenish)**
11. **Dam**

? Locate the site for a new country resort. It should have the following features:

1. **Forest land for trail rides**
2. **Water for boating and fishing**
3. **Some agricultural land for farm tours**
4. **Not susceptible to flooding**